CODE ALPHA

Empowering Future Innovators

INTERNSHIP REPORT

On

BIOINFORMATICS

Task 2: MULTIPLE SEQUENCE ALIGNMENT

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APPLICATIONS OF MACHINE LEARNING IN BIOINFORMATICS

A Mini-Project Report

1. Introduction

Bioinformatics sits at the intersection of biology, computer science, and statistics, and one of its defining challenges is scale: modern sequencing and imaging technologies generate biological data far faster than it can be manually interpreted. A single human genome contains roughly three billion base pairs, and large-scale projects now routinely produce genomic, transcriptomic, and proteomic datasets spanning thousands of individuals. Traditional rule-based and statistical methods struggle to keep pace with both the volume and the complexity of this data.

Machine learning (ML) — a set of computational techniques that allow systems to learn patterns from data rather than following explicitly programmed rules — has become one of the most transformative tools available to address this challenge. This report surveys how machine learning is applied across several major areas of bioinformatics, examines a landmark case study (AlphaFold), and discusses both the benefits and the current limitations of ML-driven approaches in biological research.

2. What Machine Learning Brings to Bioinformatics

Biological data is often high-dimensional, noisy, and only partially understood at a mechanistic level — conditions under which machine learning tends to outperform hand-crafted rules. Three broad categories of ML are widely used in the field:

- Supervised learning: Models are trained on labelled examples (e.g. sequences known to be genes vs. non-genes) and learn to predict labels for new, unseen data. Used for classification tasks such as disease diagnosis from gene expression profiles.
- Unsupervised learning: Models find structure in unlabelled data, such as grouping similar samples together. Widely used to cluster cell types from single-cell sequencing data, where the true cell identities are not known in advance.

- Deep learning: A subset of ML using multi-layered neural networks, particularly effective at learning directly from raw sequence, structural, or image data without extensive manual feature engineering. This is the approach behind most recent breakthroughs, including AlphaFold.

3. Key Application Areas

Table 1 summarizes several major domains where machine learning is now routinely applied in bioinformatics, along with representative techniques and use cases.

Table 1. Representative applications of machine learning across bioinformatics sub-fields.

Domain	Common ML Approach	Example Use Case
Sequence analysis	Recurrent neural networks (RNN, LSTM), transformers	Gene finding, splice-site prediction, promoter identification
Protein structure	Deep neural networks (e.g. AlphaFold's architecture)	Predicting 3D protein structure from amino acid sequence
Genomics / variant calling	Convolutional neural networks (CNN), random forests	Distinguishing true genetic variants from sequencing errors
Cancer genomics	Support vector machines, gradient boosting	Classifying tumour subtypes from gene expression data
Drug discovery	Graph neural networks, generative models	Predicting drug-target binding affinity, generating candidate molecules
Medical imaging	Convolutional neural networks (CNN)	Detecting tumours or lesions in MRI/CT scans
Single-cell biology	Clustering (k-means, UMAP + clustering), autoencoders	Identifying distinct cell types from single-cell RNA-seq data

A few of these areas are worth expanding on:

Sequence analysis. Genomic sequences can be treated similarly to natural language, where nucleotides or codons play the role of "words." Recurrent neural networks and, more recently, transformer-based models (the same underlying architecture used in large language models) have proven effective at tasks like identifying gene boundaries, predicting how mutations affect splicing, and locating regulatory elements such as promoters and enhancers.

Protein structure prediction. Determining a protein's 3D structure experimentally, via X-ray crystallography or cryo-electron microscopy, is slow and expensive. Deep learning models trained on the roughly 200,000 experimentally solved structures in the Protein Data Bank (PDB) have learned to predict structure directly from amino acid sequence, dramatically expanding the number of proteins whose structures can be studied (this is examined in detail in Section 4).

Drug discovery. Identifying promising drug candidates traditionally involves screening enormous chemical libraries in the lab. Machine learning models — particularly graph neural networks, which represent molecules as graphs of atoms and bonds — can predict how strongly a candidate molecule will bind to a target protein before any wet-lab experiment is run, substantially narrowing the search space and reducing cost and time.

4. Case Study: AlphaFold and the Protein Folding Problem

The relationship between an amino acid sequence and its resulting 3D structure — the "protein folding problem" — was considered one of biology's grand challenges for over 50 years. In 2020, Google DeepMind's AlphaFold system achieved a level of structure-prediction accuracy at the CASP14 (Critical Assessment of Structure Prediction) competition that was competitive with experimental methods, a result widely described as a landmark achievement for the field.

AlphaFold works by combining a deep neural network (based on the transformer / attention architecture) with evolutionary information drawn from multiple sequence alignments of related proteins across species. This mirrors, at a computational level, the biological principle explored in Task 2 of this internship: positions in a protein sequence that are conserved across species tend to be structurally and functionally important, and AlphaFold explicitly leverages this signal to constrain its structural predictions. The output includes not only a predicted structure but also per-residue confidence scores (pLDDT) and a Predicted Aligned Error (PAE) matrix — the same metrics used to interpret the human insulin structure examined in Task 3 of this internship — allowing researchers to judge which parts of a prediction are reliable.

AlphaFold's predictions for nearly all proteins catalogued in UniProt are now freely available through the AlphaFold Protein Structure Database, removing what was previously a major bottleneck for structural biologists and enabling structure-based research on proteins that would otherwise have taken years to characterize experimentally.

5. A Closer Look: How a Sequence-Based Classifier Works

To make the idea of "learning from biological data" more concrete, it is useful to walk through a simplified example: training a model to classify short DNA sequences as either a splice site or a non-splice site (a common early task in genomic ML pipelines).

- Data collection: A labelled dataset is assembled from a reference genome and its existing gene annotations — short DNA windows known to contain a splice site are labelled positive, and windows that do not are labelled negative.
- Feature representation: Raw nucleotide sequences (A, T, G, C) are converted into a numerical form the model can process — commonly one-hot encoding, where each base is represented as a small vector, or, in modern approaches, a learned embedding similar to those used for words in natural language processing.
- Model training: A neural network (or a simpler classifier such as a random forest, for smaller datasets) is trained to minimize the difference between its predicted labels and the true labels, adjusting its internal parameters over many iterations.
- Validation: The trained model is evaluated on sequences it has not seen during training, to check that it has learned generalizable patterns rather than simply memorizing the training data — a failure mode known as overfitting, which is a persistent concern in genomics because biological datasets are often small relative to the complexity of the patterns being learned.

This same general pipeline — collect labelled examples, represent them numerically, train a model, and validate on held-out data — underlies the large majority of supervised-learning applications listed in Table 1, from variant calling to cancer subtype classification. What differs between applications is mainly the nature of the input data (raw sequence, gene-expression matrix, medical image, or molecular graph) and the architecture best suited to it.

6. Benefits and Limitations

Benefits:

- Speed and scale: ML models can process millions of sequences or structures far faster than manual or purely rule-based analysis.
- Pattern discovery: Deep learning can uncover subtle, non-obvious patterns in biological data that are difficult to specify as explicit rules.
- Cost reduction: Computational prediction (e.g. of protein structure or drug binding) can substantially reduce the need for expensive, time-consuming laboratory experiments.

Limitations:

- Data dependency: ML models are only as good as their training data; biases or gaps in existing databases (e.g. underrepresentation of certain species or protein families) are inherited by the model.
- Interpretability: Deep learning models are often "black boxes" — they can produce accurate predictions without a human-readable explanation of the underlying biological mechanism.
- Confidence and error awareness: Predictions can be wrong, particularly for sequences very different from anything in the training data; well-designed confidence metrics (like AlphaFold's pLDDT) are essential but do not eliminate the risk of misplaced trust in a model's output.

7. Future Outlook

Several trends are likely to shape how machine learning is used in bioinformatics over the coming years. First, foundation models — large models pre-trained on vast amounts of unlabelled biological data (whole genomes, protein sequence databases, or single-cell atlases) and then fine-tuned for specific tasks — are beginning to appear in genomics and proteomics, following the pattern already established by large language models in natural language processing. Second, multi-omics integration is an active area of research: combining genomic, transcriptomic, proteomic, and clinical data within a single model, rather than analyzing each data type in isolation, promises a more complete picture of biological systems but raises new challenges around aligning very different types of data. Third, growing attention is being paid to model interpretability and uncertainty quantification, driven by the recognition — reinforced by AlphaFold's own confidence metrics — that a prediction is only as useful as a researcher's ability to judge how much to trust it.

For a field historically built on careful, hypothesis-driven experimentation, the growing role of large, data-driven models represents both an opportunity and a methodological shift: machine learning does not replace the biological reasoning demonstrated in tasks like sequence alignment or homology search, but increasingly automates and scales it, while still depending on domain expertise to interpret its output correctly.

8. Conclusion

Machine learning has moved from a peripheral tool to a central part of the bioinformatics toolkit, contributing to sequence analysis, protein structure prediction, genomics, drug discovery, and medical imaging. The case of AlphaFold illustrates both the scale of what is now possible — near-experimental-quality structure prediction for an entire proteome — and the underlying principle shared with more classical bioinformatics approaches: that patterns conserved across biological data, whether analyzed by a human researcher using BLAST and multiple sequence alignment or learned automatically by a neural network, tend to encode real functional and structural importance. As biological datasets continue to grow, machine learning is likely to remain one of the primary means by which that data is turned into biological understanding.

References

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